

Part I — Randomized design of (2): explicit (η, τ, ν) rates

research/formal (working)

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*Selected object = $\tilde{\theta}$ from a randomized version of constrained opt (2) only. Does **not** import Zrnic–Jordan LASSO / PoSI targets; cites their design Props. only as rate templates. Goal (iii) path: choose noise so that η (and $\tau + \nu$) can be driven small while the fixed- θ map of $\mathcal{C}$ is untouched.*

1 Setup (writeup (2))

Let $\mathcal{D}$ be calibration data with law P . Empirical maps $\hat{f}, \hat{g}$ satisfy $\sigma(\hat{f}), \sigma(\hat{g}) \subseteq \sigma(\mathcal{D})$. The deterministic program is

$$\hat{\theta}(\mathcal{D}) \in \arg \min_{\theta} \{ \hat{f}(y; \theta) : \hat{g}(\theta) \leq 0 \}. \quad (2)$$

Write $\Theta_0(\mathcal{D}) := \{ \theta : \hat{g}(\theta) \leq 0 \}$ for the (data-dependent) feasible set. When $\hat{g}$ is treated as fixed / oracle-matched on the good event below, we write Θ_0 .

Definition 1 ((η, τ) -indistinguishability — ZJ Def. 1). $Q \approx_{\eta, \tau} W$ if for every measurable O , $\mathbb{P}\{Q \in O\} \leq e^\eta \mathbb{P}\{W \in O\} + \tau$.

Definition 2 ((η, τ, ν) -stability — ZJ Def. 2). A randomized map $A : \mathcal{D} \mapsto \tilde{\theta}$ is (η, τ, ν) -stable w.r.t. P if there exists an oracle A_0 (may depend on P , not on the realized $\mathcal{D}$) such that

$$\mathbb{P}\{\omega : A(\omega) \approx_{\eta, \tau} A_0\} \geq 1 - \nu.$$

2 Assumptions on $\hat{f}, \hat{g}$

Assumption 1 (Bounded feasible diameter). On the good event of Ass. 3, Θ_0 is nonempty and $\text{diam}(\Theta_0) \leq D < \infty$ in a fixed norm $\|\cdot\|$ on the ambient space of θ .

Assumption 2 (Lipschitz objective — reserved / unused in rates). Not a hypothesis of Lemmas 1–3. Optional for future utility / sensitivity bounds: on the good event, $\theta \mapsto \hat{f}(y; \theta)$ is L -Lipschitz on Θ_0 . Do not treat L as required for the $(\eta, 0, \nu)$ rates below.

Assumption 3 (Concentration toward an oracle program). There exist f^*, g^* depending on P only (not on the realized $\mathcal{D}$) and $\varepsilon > 0$, $\nu \in (0, 1)$ such that with probability at least $1 - \nu$,

$$\|\hat{f} - f^*\|_{\infty, \Theta_0} \leq \varepsilon \quad \text{and} \quad d_H(\Theta_0(\mathcal{D}), \{\theta : g^*(\theta) \leq 0\}) \leq \omega(\varepsilon),$$

where d_H is Hausdorff distance (or a one-sided level-set inclusion) and $\omega(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$ under the instance’s regularity (e.g. continuous strictly increasing CDF). Special case (exact feasible match): $\Theta_0 = \{\theta : g^*(\theta) \leq 0\}$ pathwise — used by the nested box- $\hat{g}$ instance. Rate lemmas below are stated under the exact-match special case (or after restricting to a conservative $\Theta_0^\varepsilon \subseteq \{g^* \leq 0\}$ so the oracle feasible set is a deterministic supersystem for RNM/soft). Do not claim pathwise equality for empirical miscoverage constraints without that bridge.

Assumption 4 (Uniqueness / measurable tie-break). *For almost every noise draw used below, the noisy program has a unique minimizer (or a fixed measurable tie-break), so $A(\mathcal{D})$ is a well-defined r.v. in Θ_0 .*

3 Noisy-objective selection

Definition 3 (Noisy-objective version of (2)). *Independent of the test point, draw noise ξ and set*

$$\tilde{\theta}(\mathcal{D}; \xi) \in \arg \min_{\theta} \{ \hat{f}(y; \theta) + \langle \xi, \theta \rangle : \hat{g}(\theta) \leq 0 \}. \quad (2_{\xi})$$

3.1 Finite feasible set — report-noisy-max

Lemma 1 (RNM rates for finite Θ_0). *Assume Ass. 3–4 in the exact-match special case (or after restricting to a conservative finite $\Theta_0^{\varepsilon} \subseteq \{g^* \leq 0\}$ shared by A and A_0), and that this common feasible set is finite. For each $\theta \in \Theta_0$ draw $\xi_{\theta} \stackrel{\text{iid}}{\sim} \text{Lap}(b)$ with scale*

$$b = \frac{2\varepsilon}{\eta}, \quad \eta > 0, \quad (1)$$

and set

$$\tilde{\theta}(\mathcal{D}; \xi) = \arg \min_{\theta \in \Theta_0} (\hat{f}(y; \theta) - \xi_{\theta})$$

(report-noisy-min; equivalent to report-noisy-max on $-\hat{f}$). Let the oracle A_0 use the same ξ and f^* in place of $\hat{f}$. Then $A : \mathcal{D} \mapsto \tilde{\theta}$ is

$$(\eta, 0, \nu)\text{-stable}. \quad (2)$$

Cite: density-ratio argument as in Zrnic–Jordan Ex. 1 / stable screening (Prop. 4); sensitivity 2ε on the good event of Ass. 3. Anti-claim: RNM on a data-dependent $\{\hat{g} \leq 0\}$ with a different oracle support is not covered by this rate.

3.2 Linear Laplace perturbation (one LMO / polytope)

Lemma 2 (Linear Laplace = RNM on extreme points). *Assume Ass. 1, 3–4, that Θ_0 is a polytope, and that $\hat{f}$ is linear on Θ_0 (or that (2_{ξ}) is replaced by a single linear minimization oracle over Θ_0). Then for almost every ξ ,*

$$\arg \min_{\theta \in \Theta_0} \{ \hat{f}(y; \theta) + \langle \xi, \theta \rangle \}$$

is attained at an extreme point, so noisy (2_{ξ}) reduces to report-noisy-min over $\text{ext}(\Theta_0)$ with effective scores $\hat{f}(v) + \langle \xi, v \rangle$. Sampling ξ so that the induced extreme-point scores receive independent $\text{Lap}(b)$ perturbations with

$$b = \frac{2\varepsilon}{\eta} \quad (3)$$

(as in (1); or ξ with dual-Lap tails scaled by diameter D so that $\langle \xi, v - v' \rangle$ realizes the same score sensitivity bound on $\text{ext}(\Theta_0)$) yields the same rate as Lemma 1:

$$(\eta, 0, \nu)\text{-stability}. \quad (4)$$

*Cite: one noisy LMO step in Zrnic–Jordan Prop. 2 / Alg. 2, remapped from LASSO to general (2); density ratio as Ex. 1. For nonlinear $\hat{f}$ on a polytope, do **not** claim extreme-point attainment; use Lemma 3 or discretize to a net and apply Lemma 1.*

Remark 1 (Multi-step Frank–Wolfe composition — optional). *If $(\mathcal{Q}_\xi)$ is solved by k noisy linear minimization steps, each $(\eta_0, 0, \nu_0)$ -stable as in Lemma 2, Zrnic–Jordan composition (their Supp. Lemmas 3–4 / Prop. 2) yields either*

$$(k\eta_0, 0, k\nu_0)\text{-stability}, \quad (5)$$

$$(\tfrac{1}{2}k\eta_0^2 + \sqrt{2k \log(1/\delta)} \eta_0, \delta, k\nu_0 + \delta)\text{-stability} \quad (6)$$

(advanced composition; $\delta \in (0, 1)$ free). For goal (iii), set $\eta_0 = \eta_{\text{target}}/k$ (basic) or solve (6) for η_0 so the leading parameter equals a prescribed small η_{target} .

4 Soft near-minimizer (exponential tilt)

Definition 4 (Soft-argmin / Gibbs selection). *For inverse-temperature $\beta > 0$, draw*

$$\tilde{\theta} \sim \pi_\beta(\cdot \mid \mathcal{D}), \quad \frac{d\pi_\beta}{d\mu}(\theta) \propto \exp(-\beta \hat{f}(y; \theta)) 1_{\Theta_0}(\theta),$$

with μ a fixed reference measure on Θ_0 (counting measure if finite).

Lemma 3 (Soft-argmin $(\eta, 0, \nu)$ rate). *Under Ass. 3 in the exact-match special case (or on a common conservative $\Theta_0^\varepsilon \subseteq \{g^* \leq 0\}$), and Ass. 4 if a further argmax post-process is used, set*

$$\beta = \frac{\eta}{2\varepsilon}. \quad (7)$$

Then $\pi_\beta(\cdot \mid \mathcal{D}) \approx_{\eta, 0} \pi_\beta^$ on the good event, where π_β^* uses f^* , so soft-argmin selection is*

$$(\eta, 0, \nu)\text{-stable}. \quad (8)$$

Cite: exponential-mechanism density ratio; same role as ZJ “noisy winner” design, without PoSI. (If $\|\hat{f} - f^\|_\infty = 0$ pathwise on the common set, any β yields $(0, 0, \nu)$ -stability.)*

5 Goal (iii) reading

Corollary 1 (Small- η design). *Lemmas 1–3 make the stability parameter η a free design knob via (1), (3), or (7). Part II Cor. vanish consumes a sequence A_r with $\eta_r \rightarrow 0$ and $\tau_r + \nu_r \rightarrow 0$ at fixed classical level of $\mathcal{C}$ (no recalibration). Operationally: pick target $\eta_r \downarrow 0$, set noise scales from ε, ν of Ass. 3, and accept the utility cost (softer / noisier selection) as the price of (iii).*

Remark 2 (What is not claimed). *Rates here stabilize the **selection law** of $\tilde{\theta}$ (ZJ Def. 2). They are not leave-one-out displacement bounds for deterministic (2) (see `part_i_stability.tex`). e^η -inflation of $\mathcal{C}$ is a goal (i)–(ii) path only.*